

SHREYA NARESH MANJUCHA

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SUMMARY

Software Engineer with 3+ years of experience building user-facing applications and scalable systems using JavaScript, React, Next.js, Python, and Java. Experienced in WordPress/CMS workflows, API integration, testing, CI/CD, and collaborating across teams to deliver responsive, accessible web experience

SKILLS

Languages: Java, Python, C/C++, JavaScript, SQL

Web Development: HTML, CSS, React, Next.js, REST APIs, Git

Cloud & DevOps: AWS, Docker, Linux, Kubernetes

Databases & Data: PostgreSQL, MySQL, MongoDB, ETL/ELT

AI / ML: Machine Learning, Generative AI, Pattern Recognition, Neural Networks, Natural Language Processing, TensorFlow, PyTorch, LLM

Software Engineering: WordPress, SEO, CMS, Data Structures, SDLC, OOP, Agile, Unit Testing, CI/CD, Distributed Systems

WORK EXPERIENCE

Student Researcher — Computer Vision & Embodied AI

Aug 2025 - Present

Next-gen Efficient and Intelligent Systems (NEXIS) Lab, Purdue University | West Lafayette, IN

- Conducted AI model evaluation and optimization for egocentric video understanding using RGB-D data across 1,000+ video sequences in Linux environments, improving robustness and systematic failure analysis.
- Developed SAM2-based video segmentation pipelines achieving temporally consistent object masks in long egocentric videos, increasing segmentation accuracy by 20%.
- Built object detection and tracking pipelines maintaining persistent identities and structured 3D scene graphs over time, reducing misidentification issues by 15%.
- Evaluated models on Ego4D and Ego-Exo4D, identifying failure cases and improving overall performance by 25% through targeted analysis of occlusion, motion blur, and camera artifacts.

Software Engineer

May 2023 - Aug 2025

ICICI Lombard General Insurance Company Ltd | Mumbai, India

- Supported development and maintenance of user-facing application features through API integrations and backend orchestration.
- Improved application performance and reliability by optimizing services, reducing response time by 40% and downtime by 20%.
- Automated deployments with Git-based CI/CD workflows and strengthened testing practices to reduce release issues.
- Partnered with cross-functional teams to deliver technical solutions and ensure smooth product updates.
- Supported website and application updates through CMS-based workflows, debugging, and integration fixes, while collaborating with cross-functional teams to maintain reliable user-facing systems.

Machine Learning Intern

Jun 2021 - Jul 2021

Indian Sign Language Recognition, TSEC | Mumbai, India

- Constructed a labeled dataset of 2,000+ images and developed reproducible training and evaluation pipelines.
- Developed a real-time sign-to-text and speech system with a Python GUI supporting 2 languages, enhancing accessibility.
- Achieved 99.54% accuracy using OpenCV and scikit-learn models, significantly improving recognition performance.
- Indexed Indian Sign Language Recognition System in IEEE Xplore from presentation at ICACTA'23, increasing visibility in the research community.

PROJECTS

AI-Powered Interactive Resume Builder

May 2025 - Oct 2025

- Developed a full-stack web application with React/Next.js frontend dashboards and backend APIs for dynamic data visualization and AI-driven content generation.
- Designed REST APIs for content generation and template rendering, ensuring 100% session management reliability.
- Containerized services with Docker and automated deployments using CI/CD, reducing deployment time by 60%.
- Implemented validation checks to improve system reliability and reduce errors.
- Built the frontend using Next.js (React framework) with server-side rendering for improved performance and SEO optimization.
- Integrated data processing workflows and validation pipelines to support AI-driven personalization and recommendation features.

Automated Smart Attendance System using Face Recognition

Aug 2022 - Apr 2023

- Developed a full-stack attendance platform with a web dashboard, backend APIs, and inference services.
- Designed database schemas for identity management, attendance logs, and reporting, enhancing data accuracy by 20%.
- Achieved 98% accuracy across 100+ users in real-world deployment scenarios, improving efficiency by 30%.
- Published research on Automated Smart Attendance System in CNC'23 - Grenze International Journal, influencing industry standards.

Bhashantar - Multimodal Translation Platform

Jun 2022 - Aug 2022

- Developed a full-stack system integrating OCR, speech recognition, and NLP for seamless translation.
- Engineered backend APIs and user interface, enhancing translation functionality by 50%.
- Achieved a first-place national ranking at the Smart India Hackathon 2022, showcasing industry influence.
- Elevated system processing speed by 30%, increasing user satisfaction and engagement.

EDUCATION

MS in Computer Science: Computer Science Engineering

May 2027 (Expected)

Purdue University, West Lafayette, IN

- Relevant Coursework: Advanced Algorithms, Distributed Systems, Database Systems, Machine Learning, Data Structures, Object-Oriented Programming

